



Customer Churn and Retention Analysis for NimbusAI

Using SQL, Python, Power BI, and MongoDB

1. Introduction

In today's competitive digital landscape, understanding customer behavior and minimizing churn is critical for sustainable business growth. This project focuses on analyzing customer data from NimbusAI to identify patterns in customer retention, revenue generation, and user behavior.

The primary objective is to:

- Identify key drivers of customer churn
- Analyze revenue distribution across plans
- Understand the impact of support interactions on churn
- Provide actionable insights to improve retention

This project simulates a real-world data analytics workflow by integrating multiple tools and datasets.

2. Problem Statement

NimbusAI aims to reduce customer churn and improve revenue efficiency. However, due to fragmented data across systems, it is difficult to derive unified insights.

This project addresses:

- Why customers are churning
 - Which plans perform best
 - How support interactions impact retention
 - When churn occurs in the customer lifecycle
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3. Data Sources

3.1 Customers Table

- customer_id
- company_name

- country
- churned_at

👉 Used for identifying churn behavior and customer demographics.

3.2 Subscriptions Table

- subscription_id
- customer_id
- plan_id
- start_date

👉 Used for time-series and plan analysis.

3.3 Plans Table

- plan_id
- plan_name
- monthly_price_usd

👉 Used for revenue calculations.

3.4 Support Tickets Table

- ticket_id
- customer_id
- created_at

👉 Used to analyze customer issues and their relation to churn.

3.5 MongoDB Dataset (User Activity Logs)

- userId
- action
- timestamp

👉 Used for behavioral analysis.

4. Tools & Technologies

Tool	Purpose
PostgreSQL	Data querying and transformation
Python (Pandas, Matplotlib)	Data cleaning and analysis
Power BI	Visualization and dashboard
MongoDB	Semi-structured data analysis

5. SQL Analysis

5.1 Joins & Aggregation (Q1)

- Combined multiple tables
 - Calculated:
 - Active customers per plan
 - Revenue per plan
 - Ticket rate per customer
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5.2 Window Functions (Q2)

- Used ranking functions
 - Compared customer lifetime value across plans
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5.3 CTEs & Subqueries (Q3)

- Analyzed customer downgrade behavior
 - Correlated support tickets with churn
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5.4 Time Series Analysis (Q4)

- Used DATE_TRUNC() for monthly aggregation

- Calculated:
 - New subscriptions
 - Churn count
 - Churn rate
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Key SQL Outcomes:

- Identification of high-churn plans
 - Understanding revenue distribution
 - Detection of churn trends over time
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6. Python Analysis

6.1 Data Loading

- Imported CSV files into Pandas
 - Verified data using `.head()` and `.info()`
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6.2 Data Merging

- Merged:
 - customers + subscriptions
 - subscriptions + plans
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6.3 Feature Engineering

- Created:
 - revenue column
 - is_churned flag
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6.4 KPI Calculation

- Total Revenue
- Churn Rate

- Plan-wise revenue
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6.5 Visualization

- Bar charts for revenue distribution
 - Churn comparison graphs
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7. Power BI Dashboard

7.1 KPIs

- Total Customers
 - Total Revenue
 - Churn Rate
 - Total Tickets
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7.2 Visualizations

Revenue by Plan

- Shows top-performing plans

Customer Distribution

- Identifies plan popularity

Churn Analysis

- Highlights high-risk plans

Ticket Analysis

- Links support issues with churn

Time Series Chart

- Shows growth and churn trends
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7.3 Dashboard Features

- Interactive slicers (country, plan)
- Drill-down capability

- Clean layout for storytelling
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8. MongoDB Analysis

8.1 Data Exploration

- Used find() to preview data
 - Used countDocuments()
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8.2 Aggregation Queries

- Grouped by action:
 - Most frequent activities
 - Grouped by user:
 - Most active users
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8.3 Data Issues Identified

- Inconsistent naming (customer_id vs customerId)
 - Missing values
 - Mixed data types
 - Schema inconsistencies
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9. Key Insights

Insight 1:

Customers with more support tickets show higher churn rates

👉 Indicates service quality impacts retention

Insight 2:

Premium plans generate higher revenue and lower churn

👉 High-value customers are more stable

Insight 3:

Churn is highest in early customer lifecycle

👉 First 1–2 months are critical

10. Challenges Faced

- SQL syntax errors in complex queries
 - Handling inconsistent data across systems
 - MongoDB .js file import issues
 - Mapping data across multiple tables
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11. Data Quality Issues

- Missing values
 - Duplicate records
 - Data type mismatches
 - Timezone inconsistencies
 - Schema mismatch (SQL vs MongoDB)
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12. Conclusion

This project demonstrates an end-to-end data analytics pipeline. By integrating SQL, Python, Power BI, and MongoDB, meaningful business insights were derived. The findings highlight the importance of customer experience, pricing strategy, and early engagement in reducing churn.

13. Future Scope

- Machine learning-based churn prediction
 - Cohort analysis for retention tracking
 - Real-time dashboard integration
 - Automated ETL pipelines
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14. Author

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